Guarded resultants and early projection for Diophantine consistency

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Abstract

We construct explicit ordinary polynomials whose lack of solutions in specified integer domains is provably equivalent in ZFC to syntactic consistency of ZFC. A guarded quadratic resultant removes a signed strong trace and yields thirteen-integer equations of degrees 118 and 120. An early Pell projector excludes the third main index before sign recovery, licensing a squared transport and a ten-positive-variable degree-190 equation. Natural-domain coordinate charts and certified contractions produce a 201-variable quartic and degree-six and degree-eight alternatives. An even affine horizon gives a 209-integer quartic, trading one variable and one residual against support and coefficient height. Every result includes complete coefficients or circuits, constructive maps, ordinary proofs and reproducible finite checks. The pinned low-padding2963 specializations retain their own whole native calibration. No consistency assertion, global optimum or new end-to-end kernel certificate is claimed.

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1 Objective and the comparison boundary

For each specified domain D and fixed polynomial P, the target is

$$\mathrm{ZFC} \vdash \left(\neg\exists \boldsymbol{x} \in D^n \; P(\boldsymbol{x}) = 0\right) \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}).$$

The proofs establish solvability equivalent to a finite syntactic refutation. They do not assert consistency, unsolvability, or an actual solution. The domains $\mathbb{Z}$, $\mathbb{N} = \{0, 1, 2, \ldots\}$ and $\mathbb{Z}_{>0}$ are part of the individual theorems, not interchangeable size conventions.

The immutable source snapshot of ProveIt is

`ecb9a4b4dda617512952525d50f17343700aac52`

It includes the newer reciprocal-pooling, scaled-center and CRT-index-anchor constructions, as well as preceding reports. The accompanying baseline audit checks complete polynomial schemas, not just convenient file names or headline tables. In particular, an incoming 209-variable system has an additional reciprocal residual: its single polynomial has degree 24, not degree four. Omitting that residual would give a false comparison.

The sole bounded late comparison is commit

`c8187221a60df5baa0047c254a592a014491b60c`

Its incoming Pell-growth210 equation superseded our preliminary 212-variable scaled-pair numerical endpoint. The final 209-variable construction below modifies that stronger source with a separately proved even affine horizon. The 212-variable packet is retained only as a verified method control. The later block-mask page uses exactly the same four 2963 integers and specialized polynomial; its printing improvements are not a new tuple or a change to our frozen calibration. No later source snapshot is used for the claims in this release.

Measurements are the complete number of independent quantified coordinates, exact degree, residual count, collected support, maximum absolute coefficient and live binary operation count. Circuit counts charge every addition, subtraction and multiplication, including squares and scalar products; reuse and integer leaves are free. They are not bit complexity or a model with free rational constants. Fixed ports are not variables, but their entire coherent numerical tuple and its calibration are specified.

Domain	Variables	Degree	Operations	Matched comparison
$\mathbb{Z}$	13	118	114	Earlier 152/119 and 156/117
$\mathbb{Z}$	13	120	112	Earlier 152/119 and 156/117
$\mathbb{Z}_{>0}$	10	202	97	Earlier 206/90
$\mathbb{Z}_{>0}$	10	198	111	Degree/cost tradeoff
$\mathbb{Z}_{>0}$	10	190	119	New squared-transport route

Table 1: Complete integer-leaf circuits. Comparison pairs are degree/operations with the domain and variable count held fixed.

2 Results and tradeoffs

Both thirteen-integer points improve degree and operation count against both matched predecessors. They trade degree against cost with one another. The ten-positive-variable points lower degree at increased operation cost; they do not uniformly dominate the earlier 90-operation degree-206 construction. Domains and full coefficient tuples remain separate.

Domain	Variables	Degree	Rows	Terms	Height
$\mathbb{N}$	201	4	153	6,002	1,860,103
$\mathbb{N}$	165	6	117	6,558	1,860,101
$\mathbb{N}$	164	6	116	15,774	1,860,101
$\mathbb{N}$	139	8	91	11,873	1,117,699,986
$\mathbb{N}$	138	8	90	16,515	2,933,403,776
$\mathbb{Z}$	209	4	149	5,486	17,592,186,044,416

Table 2: Complete direct polynomials. Height is the largest absolute collected coefficient; higher-degree rows are not called quadratic.

The natural quartic reduces the previous 202-variable, 154-row, 9,353-term equation to 201 variables and 6,002 terms, with a small height increase from 1,854,167 to 1,860,103. The older 202-variable height-416,488 equation remains a lower-height alternative. The degree-six and degree-eight results exchange degree for fewer variables; none is represented as a quartic.

The 209-integer quartic has height 2^{44} and largest residual coefficient 2^{22}. Against the later-inspected 210-variable, 150-row equation with 5,282 terms and height 2^{40}, it saves one variable and one row but adds 204 terms and multiplies height by 16. This is a dimension tradeoff, not uniform dominance. The earlier 215-variable low-height equations remain separate alternatives.

The preliminary 212-variable construction is preserved as a method control in the archive, with its distinct scaled-Pell proof. It is numerically dominated by the incoming 210 and is not a final frontier claim. The tempting odd-affine horizon was rejected because of a signed $t = -1, x = 0, q = 1$ bootstrap escape; the accepted horizon $t = 4u + 2$ explicitly excludes both 0 and -1 before sign recovery.

The pinned low-padding2963 coefficient tuple uses 69,685 raw bytes. Its encoding is credited to the incoming source. The present work verifies its complete binding to each new positive front rather than inventing that encoding or silently changing a previously calibrated polynomial.

3 Guarded quadratic trace elimination

Two new fixed ordinary polynomials have thirteen unrestricted integer unknowns:

$$(\deg, \mathrm{ops}) = (118, 114), \qquad (120, 112).$$

The charged operation counts (multiplication, addition, subtraction) are $(66, 27, 21)$ and $(65, 26, 21)$. Each point strictly dominates the incoming thirteen-integer $(152, 119)$ and $(156, 117)$ points in both varying metrics; the two new points themselves trade degree against cost. All six full fixed numeral payloads and their finite recipe are retained unchanged. These are ordinary reductions through an identified native/compiler/refutation theorem, not a consistency assertion, global optimum or new end-to-end kernel certificate.

3.1 Entire literal equations

The inputs, in order, are

$$j, C, z, h, r, v, w, T, c, n, y, p_2, p_3 \in \mathbb{Z}.$$

Use the authentic ports $\beta, K_0, M_C, M_F, A_c, B_c$, with M_F already shifted, $\beta = 2^{1064921}-1$, $A_c = 2^{1113}$ and $B_c = 2^{666104}$. Their full bytes, binding and finite recipe are under the retained coefficient directory:

`roots/inherited_round22/inherited_round20/`
`inherited/coefficients/`

All following names are polynomial abbreviations, not further unknowns:

$$J = 6j + 1, \quad q_m = \beta J, \quad q = q_m + 1, \quad W = 12w + 4, \quad X = qW, \quad Y = v,$$
$$F = (K_0 + W)C - zq_m, \qquad R = [q(q - F) - C](q^2 - 1) + (M_C + qM_F)J,$$
$$E = XY, \quad a = E + Y, \quad H = 4a + 3, \quad \Delta = a^2 + H, \quad \ell = XY^2,$$
$$\eta = c - Yk, \quad L = \ell k, \quad I = (8r + 4)c^2, \quad D_s = \Delta I^2, \quad F_s = D_s + 1,$$
$$Q = \Delta D_s, \quad \delta = Q - 1, \quad b = cT - R, \quad d_y = \delta y, \quad t = c + d_y, \quad t_t = t^2, \quad L_b = F_s b^2,$$
$$A_r = L_b + t_t - Qd_y y - 1, \qquad H_r = A_r^2 - 4L_b t_t,$$
$$q_3 = q^3, \quad M = 2q_3, \quad U = MH, \quad K_{\mathrm{lo}} = X(2X - 3c) - 2,$$
$$Z_0 = Un + MK_{\mathrm{lo}} + H(Y - q_3), \quad D_{\mathrm{err}} = (A_c F - q)^2 + (B_c C - q)^2.$$

The two charts are

$$k_{118} = R + 1 + h\ell, \qquad \mathcal{B}_{118} = (Yk)^2(q^2 - D_{\mathrm{err}}) - q^2[4(Y\eta)^2 + (kq^2)^2], \qquad Z_{118} = Z_0 + UL,$$
$$k_{120} = R + 1 + hE, \qquad \mathcal{B}_{120} = (Yk)^2(q^2 - D_{\mathrm{err}}) - q^2[4(Y\eta)^2 + (kq^3)^2], \qquad Z_{120} = Z_0.$$

In the 120-chart evaluator, $L = E(Yk)$ avoids a separate computation of ℓ. Finish in either chart with

$$G = \mathcal{B} - 1 - p_2^2 - p_3(p_3 + 1), \quad D_\tau = 1 + L(L + k), \quad U_2 = U^2, \quad g_a = U_2 G, \quad d_b = U_2 D_\tau,$$
$$N = (Z^2 - g_a - d_b)^2 - 4g_a d_b, \qquad P = NH_r + N + H_r = (N + 1)(H_r + 1) - 1.$$

The expanded final schedule saves one operation. There is no division gate, variable exponent, omitted numeral or hidden positivity restriction.

3.2 Resultant identity and the exceptional branch

For proof only, introduce f and put $V = fb - c$ and $E_a = V^2 - \delta y(2V + y)$. The complete identities are

$$A_r = F_s b^2 + c^2 - 1 - \delta y(y - 2c), \tag{1}$$

$$E_a - 1 = A_r - 2btf + b^2(f^2 - F_s), \tag{2}$$

$$H_r = A_r^2 - F_s(2bt)^2. \tag{3}$$

Thus $f^2 = F_s$ and $E_a = 1$ imply $H_r = 0$. The converse requires $bt \neq 0$; without it a resultant may vanish without a rational trace. We establish that nonvanishing from the guard *before* recovering f. Sharing t_t in the first and third expressions saves two evaluator gates; identity (1) must be used before assigning degrees.

3.3 All signed roots and guard restoration

Unconditionally q is even and nonzero, $|W| \geq 4$, $|X| \geq 8$, H is odd and nonzero, and $U \neq 0$. Since $a = (X+1)Y$ is zero or has $|a| \geq 7$, $\Delta = (a+2)^2 - 1 \geq 3$. Also ℓ is even and $D_\tau = 1 + \ell(\ell+1)k^2 \geq 1$.

Both N and H_r are squares modulo four. At $P = 0$, the integer factors $N + 1, H_r + 1$ are units, neither can be -1, and consequently $N = H_r = 0$. With $\rho = Z/U$,

$$(\rho^2 - G - D_\tau)^2 = 4GD_\tau.$$

This forces $G \geq 0$, including $G = 0$, and $\mathcal{B} > 0$, hence $Yk \neq 0$. Moreover

$$D_\tau \geq 1 + \tfrac{1}{2}X^2Y^4k^2 \geq 1 + 8q^2Y^4k^2 > q^2Y^2k^2 \geq \mathcal{B} > G.$$

Therefore $\rho \neq 0$ before any division by it. The rationals

$$p = \frac{\rho^2 + G - D_\tau}{2\rho}, \qquad \tau = \rho - p$$

satisfy $p^2 = G$ and $\tau^2 = D_\tau$. A rational whose square is integral is integral, by reducing its fraction to lowest terms. Thus $p, \tau \in \mathbb{Z}$. This handles zero guard roots, both signs and coincident square classes.

Set $\sigma = 1$ for degree 118 and $\sigma = 0$ for degree 120, and restore

$$p_{1,\mathrm{old}} = p, \qquad n_{\mathrm{old}} = n + \sigma L - (2L + k) - p + \tau.$$

With $Z_{\mathrm{old}} = Un_{\mathrm{old}} + MK_{\mathrm{lo}} + H(Y - q^3)$, one has $Z_{\mathrm{old}}/U + 2L + k = 2\tau$. Hence the identity

$$(Z_{\mathrm{old}} + Uk)^2 + 4U(LZ_{\mathrm{old}} - U) = (Z_{\mathrm{old}} + U(2L + k))^2 - 4U^2D_\tau$$

restores the old carrier row, while $p^2 = G$ restores its guard.

3.4 Noncircular exclusion of every resultant escape

Only the positive budget and the retained centered-disk/authentic-mask argument are now used. Write $q_0 = |q|$ and $B_0 = k^2(q^2 - D_{\mathrm{err}}) - 4q^2\eta^2$. The 118 guard is $Y^2B_0 - k^2q^6 > 0$, so $B_0 > 0$ and $|Y| > q_0^2$; the 120 guard similarly gives $|Y| > q_0^3$. The disk gives $D_{\mathrm{err}} < q^2$, $|\eta/k| < 1/2$, and

$$|c| > |k|(|Y| - 1/2).$$

The authentic mask lemma gives, independently of the strong or auxiliary rows,

$$q_0 \ge 16, \qquad q_0(q_0^2-1) < R < q_0^4 - 2.$$

Thus $|\ell| > 4q_0^5 > 3R+1$ in the 118 chart, while $|E| > 4q_0^4 > 3R+1$ in its companion. These are the early arguments in the retained source 153 Section 2 and the q8 signed source, before any Pell-index interpretation.

Let $d = \ell$ or E. Since $k = R+1+hd$ and $|d| > 3R+1$, either $h = 0$ and $|k| = R+1$, or $|k| \ge |d| - (R+1) > 2R \ge R+1$. Consequently $|c| > R > 0$. For integer T, $b = cT - R$ cannot vanish: $T = 0$ gives $-R$, while $T \ne 0$ gives $|cT| \ge |c| > R$.

Also $|I| \ge 4c^2$ and $\Delta \ge 3$, whence

$$\delta = \Delta^2 I^2 - 1 \ge 144c^4 - 1 > |c| > 0.$$

For every integer y, $t = c + \delta y \ne 0$: this follows from $c \ne 0$ if $y = 0$ and from $|\delta y| > |c|$ otherwise. Therefore $2bt \ne 0$. Equation (3) now gives a rational

$$f = \frac{A_r}{2bt}, \qquad f^2 = F_s \in \mathbb{Z},$$

so f is an integer. Equation (2) gives $E_a = 1$, and $f^2 - \Delta I^2 = 1$ is immediate. Arbitrary signs of every input remain allowed. The cases $c = 0$, $b = 0$ and $t = 0$ were excluded by prior inequalities, not silently divided away.

We have restored a literal zero of the even-strong fifteen-integer kernel identified by equation (2.5) of the retained Round 22 proof. Its Sections 3–4 supply the complete signed bootstrap and *fresh* odd-scale strong and auxiliary witnesses before invoking the authenticated native/compiler/refutation theorem. We do not incorrectly reuse an even-scale trace in that odd-scale theorem.

3.5 Reverse map and constructive completeness

From any zero of the even-strong kernel, $f^2 = F_s$ and $E_a = 1$ imply $H_r = 0$ without division. Its carrier gives $(Z_{\mathrm{old}}/U + 2L + k)^2 = 4D_\tau$. The rational-root argument makes this quantity an even integer, say 2τ. Delete $f, p_{1,\mathrm{old}}$ and set

$$n = n_{\mathrm{old}} - \sigma L + (2L+k) + p_{1,\mathrm{old}} - \tau.$$

Then $\rho = p_{1,\mathrm{old}} + \tau$, $G = p_{1,\mathrm{old}}^2$, and $D_\tau = \tau^2$, so $N = 0$ and $P = 0$. This works on every old zero with all root signs; no inverse polynomial automorphism is claimed. The retained fresh-witness construction and native completeness theorem therefore give a new zero from every finite syntactic ZFC refutation. Together the two maps establish the ordinary ZFC-provable equivalence of integer unsolvability and syntactic Con(ZFC).

3.6 Complete exact leading forms and verification

Write

$$q_0 = 6\beta j, \quad X_0 = 12q_0 w, \quad a_0 = X_0 v, \quad F_0 = 12wC - q_0 z,$$
$$R_0 = -q_0^3 F_0, \quad \ell_0 = X_0 v^2, \quad U_0 = 8q_0^3 a_0.$$

Identity (1) gives $\deg A_r = 22$, not 38. The whole resultant top is

$$[H_r]_{60} = -2^{20} a_0^{10} r^6 c^{12} R_0^2 y^2.$$

For degree 120,

$$[N]_{60} = U_0^4 R_0^4 (q_0^8 + \ell_0^2)^2.$$

For degree 118 put $k_0 = R_0 + h\ell_0$ and $T_0 = A_c^2 v^2 F_0^2 + 4q_0^2 v^4 + q_0^6$. The exact cancellation

$$(Z_0 + UL)^2 - U^2 D_\tau = Z_0^2 + 2ULZ_0 - U^2(1 + Lk)$$

has degree 26, and gives

$$[N]_{58} = 4U_0^4 \ell_0^2 k_0^4 T_0.$$

The full output tops are the products of these forms. Their 85 and 21 monomials are all expanded by the independent checker. In particular,

$$[j^{58} w^{16} v^{18} z^6 r^6 c^{12} y^2] P_{118} = -2^{124} 3^{74} \beta^{58} \neq 0,$$
$$[j^{66} w^{14} v^{14} z^6 r^6 c^{12} y^2] P_{120} = -2^{126} 3^{80} \beta^{66} \neq 0.$$

The exponents sum to 118 and 120. These are coefficients at the actual fixed numeral slice, not a generic-coefficient estimate or sampled ray.

Required replay uses only Python's standard library. The independent checker imports no producer or circuit implementation, binds every live gate to a separately written formula, checks all signed identities and leading forms, authenticates 255 inherited files and all 615,883 raw coefficient bytes, and rejects both alternative binary operators at each live gate (228 and 224 mutations). Normal and optimized runs have identical explicit acceptance conditions. The separate finite recipe reconstructs all six numeral payloads. Finite checks verify literal artifacts and algebra; the unbounded guarded extraction and inherited logical calibration remain the stated ordinary proofs.

4 Quotient centers and early projection in the free ten-positive chart

4.1 Result, domain, and attribution

This package gives three ordinary polynomials in exactly ten independently quantified strictly positive integer variables:

Graph prefix	Operations	Exact degree	Transport	Root scale
quotient_t	97	202	bV	2
product2	111	198	bV	2
product4	119	190	b^2	4

Every graph is supplied at two separate complete native/refutation calibrations: historical indexed943/radix67 and pinned low-padding2963. The latter has 69,685 raw coefficient bytes, not the frozen4177 tuple's 98,219 bytes. Each of its four integers, native proof, exact942-selector bijection, physical marker and Pclp controller changes as one coherent unit. No coefficient port, old word or controller certificate is transferred alone.

The free chart Y=s, lambda=q, inverse-Lucas correction and unsquared trace are credited to source172, whose full retained parent is `parent/`. Source178 supplies the A-quotient idea and the full-product Omega4 evaluator; its original proof and algebra are retained in `inherited/`. Those incoming scaled-chart profiles 96/220 and116/208 do not improve the prior free-chart 90/206. Our

contributions are the compatible free-chart composition, the cheaper t=A-1 quotient, first-product truncation, and early p=3 projector exclusion which newly licenses square transport here. The last step avoids illicitly importing the scaled-chart q|Y argument.

All binary +,-,* gates, squares and scalar products are charged. Fixed integer leaves and reuse are free. These are measured tradeoffs, not a global optimality or literature-priority claim. Ordinary soundness/completeness, composed with the identified retained syntactic calibration, proves in ZFC:

```
The equation has no positive-integer zero iff Con(ZFC).
```

No consistency decision, actual zero, unsolvability assertion, new compiled formal-kernel replay or new encoded internal-ZFC proof object is claimed. Finite algebra and byte checks do not prove the unbounded lemmas by testing.

4.2 The complete ordinary polynomials

The independent variables in graph order are

```
J,F,alpha,i,T,w,s,n,eta,zeta > 0.
```

Fix the entire chosen tuple beta,K0,MC,MF. Both actual tuples satisfy beta>=73 odd, K0>0 and 0<MC<beta<MF<2beta. The strengthened beta bound is used explicitly in section3; no theorem for arbitrary beta>=31 is asserted. MF is the source mask with beta added exactly once, not the native mask. Define these polynomial abbreviations, none a new quantified variable:

```
h=beta J, q=h+1, C=q-F-alpha, X=qw, Y=s, Ybar=qY,
k=eta+zeta, c=kY+eta,
A=Ybar(X+1)+1, Delta=A^2-1, H=4A-5,
v=X Ybar^2, ell=vk, E=1+ell(ell+k),
R=[h(q-F)+alpha](q^2-1)+(MC+qMF)J,
B=(K0+w)C-F, U=ic^2, a=Delta U^2E, F0=1+a,
G=EF0, f=a+F0, Q=4Delta aF0,
b=cT-R, V=fb-c, K=X(3c-2X)+2,
e6=6(k-R-1)-ell(k^2-4), D=6q^7, M=hDH.
```

Choose a row below; the formulas are division-free integer polynomials. For quotient_t put

```
Qt=(kYbar)[ell+k-(kYbar)Ybar],
C2=U(2AE-Qt), Omega=C2 V, r=2, L=bV.
```

For product2 put

```
S=(kYbar)[ell+k-(kYbar)(Ybar+1)], theta=U^2b,
Omega=U{2A[EV-theta(E^2+S^2)]
              +S[4theta E-(b-c)]}, r=2, L=bV.
```

For product4 put

```
S=(kYbar)[ell+k-(kYbar)(Ybar+1)], theta=U^2b,
e2=(kYbar)^2, Rc=theta(3E+e2)-(b-c),
Sc=4E^2+3S^2+e2^2,
Omega=U[2S Rc+A(4EV-theta Sc)], r=4, L=b^2.
```

In every case set

```
Z0=hD[H(n-LB)+K]-H[DB+h(e6-6q^4s)],
Z=Z0-M Omega,
P=Q(Z^2-r^2M^2GV^2)+(r^2M^2G-Z^2).                    (1)
```

The single equation is P=0. No rational constant, division, recurrence, radical, floor, variable exponent, hidden witness or oracle is a graph gate. The quotient_t implementation shares V in n-V(bB+C2) and computes 2A=(A-1)+(A+1) using two already-live registers.

For proof only put t0=B/h, kappa=K/H, rho=(e6/6)/q^7, sigma=Y/q^3, W=1+hL and z=Z/M. Before integrality is established these are rational quantities. Identically,

```
z=n+kappa+sigma-Wt0-rho-Omega,
P=(Q-1)Z^2-r^2M^2G(QV^2-1).                          (2)
```

4.3 Root extraction before sign or denominator assumptions

At every positive input q>=74, X>=q, Y>=1, c>=3, A>q^2, H>0, U>=c^2, f>2c and Q>c^2. The exact identities

```
f^2-1=4aF0, Q=Delta(f^2-1)=4Delta^2U^2G,
gcd(E,F0)=1, f-1=2Delta i^2c^4E
```

are unchanged from the retained parent. V cannot vanish. If P=0, the integer equation(2), valuations at primes dividing G, and the square factor r^2 show that G is a square. Thus E=tau^2, F0=f0^2 and g=tau f0>0. The parent's rational adjacent-Pell descent, unchanged by the constant square scale, gives y=|z|/(rg) an integer and

```
QV^2-(Q-1)y^2=1, z=epsilon rgy, epsilon=+/-1.
```

In particular z is integral. The signed all-index folding theorem applies with I=2Ug, f^2-Delta I^2=1, Q=(Delta I)^2 and c^2|I. It gives c=psi_A(p) for p positive odd; c>=3 excludes p=1. Hence

```
p>=3, c>=c3=4Delta+3>4q^4, K>0.
```

All of this is independent of the signs of R,B,C,b,V. No q|Y, authentication p=R, auxiliary-index-three exclusion or large-rank ratio bound has been used.

The first square gives k=2psi_(1+2v)(m), m>=1. m=1 gives k=2, eta=zeta=1 and c=2Y+1 strictly between the first two main ordinates, so m>=2. The first base 1+2v>A, and c>k. Strict base/index monotonicity therefore gives m<p: if m>=p, psi_(1+2v)(m)>psi_A(p)=c, impossible. This m<p fact does NOT use p=R or any mask bound.

4.4 New early projector exclusion of main index three

Integrality of z already gives H|Z. Reducing the literal numerator(1) modulo H gives H|hDK. Since D=6q^7 and gcd(H,q)=1,

```
H | 6hK.                                             (3)
```

The shift M Omega and either transport are multiples of H; they do not change this step. We retain the entire factor6h.

Suppose p=3. Then m=2 by section2. Write u=X+2-2Y. Using c=4Aˆ2-1, k=8Xqˆ2Yˆ2+4, eta=c-kY gives the exact polynomial identity

```
eta=4qˆ2Yˆ2(Xu+1)+8qY(X+1)-4Y+3.                (4)
```

If $u \le -1$, the first term is at most $-4q^2Y^2(X-1)$. Since $qY \ge 32$ and $X \ge 32$, we have $qY(X-1) > 2(X+1)+1$, so expression (4) is negative. If $u \ge 2$, subtracting k leaves at least

$$4q^2Y^2 + 8qY(X+1) - 4Y - 1 > 0.$$

Both alternatives contradict $0 < \eta < k$. Thus u is 0 or 1, so $2Y = X+2$ or $X+1$, and

```
H=4qY(X+1)-1>2qXˆ2.                             (5)
```

For p=3 the projector is the direct identity

```
4K+(X-8)(8X+1)=3XH(4A+5).                       (6)
```

Together(3),(6) show H divides the positive integer N=6(q-1)(X-8)(8X+1). Because X>=32, 0<N<48qXˆ2. By(5), j=N/H is an integer with1<=j<=23. Modulo q, H congruent -1, X congruent 0 and N congruent 48, so j congruent -48 modq. But49<=j+48<=71<q, impossible. Therefore p>=5 and c>c3. This is a universal proof at the explicitly verified beta>=73 boundary; finite tests of small inputs are not its justification.

Now b cannot vanish by the adjacent-square gap argument from the parent. It has the sign of V. The auxiliary Pell index is odd and at least3. At index3 its two signs would give respectively

```
f(b-4Delta f)=c-c3,
f(b+4Delta f)=c+c3.
```

Both right sides lie strictly between0 and2c<f, impossible. Thus that index is at least5, and

```
|V|>=16Qˆ2-20Q+5>8Qˆ2,
|b|>8fˆ3-1/2>7fˆ3, f>6g,
y<3|V|/2, |V|<3f|b|/2.                          (7)
```

This finally licenses square transport. The proof did not silently exclude an index-three branch before its required early projector argument. For the two bV rows the weaker inherited bound |b|>4Delta f-1 would already suffice: Delta>=4 and f>6g imply 4Delta f-1>96g-1>72g for integer g>=1. their validity does not depend on square transport.

4.5 Centers are small, with both signs covered

Let C0=AUE. Since G=E+Delta Uˆ2Eˆ2,

```
0<C0<3g/2.                                      (8)
```

For quotient_t set Rt=1+kˆ2Ybarˆ2(Ybarˆ2-1)>0. Exact polynomial division by t=A-1 gives E=(A-1)Qt+Rt. Also Qt=kˆ2Ybar[(X-1)Ybarˆ2+1]>0, hence Qt<E/(A-1). Consequently 0<C2=U(2AE-Qt)<2AUE. Thus Omega has the sign of V and

```
2gy+|Omega|<3g|V|+3g|V|=6g|V|<L/2.          (9)
```

For the product rows, temporarily regard A as an indeterminate T over Z[Ybar,k,U,b,c]. Then

```
E(T)=e2 T^2+e1 T+e0,
e2=k^2Ybar^2,
e1=k^2Ybar(1-2Ybar-2Ybar^2),
e0=1+k^2Ybar(Ybar+1)[Ybar(Ybar+1)-1],
V(T)=(1+2(T^2-1)U^2E(T))b-c, F(T)=E(T)V(T).
```

Let Qj be the value at A of the polynomial quotient of F(T) by T^j. Then exactly

```
Omega2=U(2AF-Q1),
Omega4=U(4AF-2Q1-Q3/2).                    (10)
```

Q3 is coefficientwise even, because the only odd-coefficient candidate (b-c)E has T-degree at most2. Equations(10) are proof-only descriptions; the literal evaluators in section1 use no quotient or division gate.

For a polynomial, superscript # denotes absolute-coefficient evaluation at A. Here $\overline{Y} = qY$ denotes the register `Ybar`. Put $\varepsilon = (\overline{Y}+1)/A < 1/16$. The coefficient estimates

$$|e_1| < 2e_2(\overline{Y}+1), \qquad 0 < e_0 \le e_2(\overline{Y}+1)^2$$

give $E^{\#}(A) < 2E$. Moreover,

$$c+1 < 2U^2E, \qquad |V| > (f-c)|b| > \Delta U^2 E|b|.$$

Consequently,

```
F#(A)<2E(|b|+c)+8(A^2+1)U^2E^2|b|,
F#(A)/|EV|<8+20/Delta<10.
```

Deletion of low coefficients gives |Qj|<=F#(A)/A^j. As A>=16,

```
|Omega2-2AUEV|<AUE|V|,
|Omega4-4AUEV|<AUE|V|.
```

Therefore both have the sign of V, and

```
|Omega2|<3AUE|V|,
3AUE|V|<|Omega4|<5AUE|V|.                  (11)
```

For product2 with L=bV, (8),(11) and |b|>72g yield 2gy+|Omega2|<(15/2)g|V|<L/2. For product4 with L=b^2, (7),(8),(11) give

```
4gy+|Omega4|<(9/4)f|V|
  <(27/8)f^2|b|<b^2/4.                      (12)
```

The last inequality follows from |b|>7f^3 and f>2. All bounds precede sign recovery and cover negative b,V. We do not call Omega positive until V itself has been proved positive.

4.6 Signed bootstrap, authentication, and projection order

The inverse correction satisfies e*=e6/6<=k-R-1<b for every integer R, so rho<|b|/q^7. In both weights L>0 and L>=|b|. If B<=-1, Wt0=B/h+LB<-L. Solving(2) for positive n and using(9)–(12), positive kappa,sigma, and q^7>2 gives

```
n< L/2-L+|b|/q^7<0,
```

a contradiction. Thus B>=0, and (K0+w)C=B+F>0 gives C>0. Only now use the selected whole tuple's rank bound3q<R<q^4-2. As c>4q^4>2R, b,V are positive. The unchanged normalized main-index and signed auxiliary-index theorems applied to V congruent -c modf and V congruent -R modc identify p=R. Mask parity now gives q even,J odd.

Every Omega is an integer polynomial. Its numerator contribution M Omega is divisible by D and H, so all subsequent modular reductions are exactly those of parent/PROOF.md sections4–5. We state their noncircular order. First k=2psi_(1+2v)(m) is even and e* is integral. Literal cancellation of6 in the rational carrier gives q^4|e*. The full Lucas identity is

```
k=2m+8v Cm+32v^2Zm, Cm=binom(m+1,3), Zm>=0,
3[e*-(2m-R-1)]=4v^2[24Zm-Am Sm],
Am=Cm+4vZm, Sm=k^2+k(2m)+(2m)^2-4.
```

With v=q^3ws^2 the latter gives q^4|e*-(2m-R-1) at every prime, including3. Since m<R and R<q^4, m=(R+1)/2. One does not yet infer q^7|e*. Write R=2r0+1; r0>=48. The lambda-uniform ratio theorem at lambda=q,Y>=1 gives Y>X^r0/3 and X^(r0+1)<H. The full-product projector bound, retaining6h, gives X=2^R. Hence q is a power of2; rank growth supplies q^6|X and v is divisible by q^8. Only now does the cubic identity imply q^7|e*. Thus rho and kappa are integral.

The remaining carrier relation yields q^3|Y and h|B, since W congruent 1 modh. Therefore t0 is a nonnegative integer. The retained fixed-minus parity, lambda-uniform half-binomial rounding and fresh native reconstruction at A_source=Y(X+1)+2 produce actual source words for the selected whole calibration. This base is not confused with A=qY(X+1)+1. The native interpretation yields a finite syntactic ZFC refutation, not merely a conclusion from elementary scalar boxes.

4.7 Constructive strictly positive completeness

Conversely, the chosen whole native/refutation calibration supplies fresh canonical data q=2^a, X=2^R, R congruent 3 mod4,3q+1<=R<q^4,q^3|Y,C>0 and t0=B/h>=0 integral. Parent section6 is uniform in these data and constructs the free-chart inputs w=X/q,s=Y, positive eta,zeta,i,T, with c=psi_A(R), k=2psi_(1+2v)((R+1)/2), g>0,V>0,y>0 and

```
g>kappa+sigma+|rho|.
```

No old packed word or strong period is reused across calibrations. Every center is positive when V>0 by section4. Set, at the selected transport and root scale,

```
n=rgy+Omega+(1+hL)t0+rho-kappa-sigma.          (13)
```

All terms are integers and the preceding strict bound makes n>0. The literal numerator satisfies z=rgy, so(1) vanishes. This construction also handles possibly negative rho. For bV one may equivalently take n_new=n_parent+(r-1)gy+Omega. For b^2 we use(13), not a claim that subtracting bVB from an arbitrary positive parent n preserves positivity. All ten independent coordinates have their declared strict positive domain.

4.8 Entire low-padding2963 native calibration

The new native proof, decoder, language and every historical ordinary interface it names are retained in native2963/, with267 exact source pins. Its complete proof is research/low-padding-independent/PROOF.md. The admissible selector bijection g(0)=0,g(j)=j+1 (1<=j<=941) removes exactly old forbidden selector1. It preserves and reflects the full language, including the complete436-character seed,430 actions, renamed initializer928 and calibrated final tile941; it is not an unproved canonical subfamily.

There are3297 coefficient slots,976 weighted positions and978 physical positions. Physical dummy977 and marker1005 have zero coefficient but remain Boolean positions. A separate kernel monomial copies Start0 to1005. All916 padding tests use bit2 of fields1..916. The sole clause is1956; marker parity, Boolean temporal count, alignment, carries and reset tests are proved before the resulting native interpretation. Soundness needs no controller premise. Completeness uses the same tuple's discharged Pclp:

```
rho_native=2^47, L_native=2963, d=139261,
fixed prime factors47,2963, orders23,2962,
spacing68126, auxiliary prime1362521,
H residues29,2938,492699.
```

The finite checker reconstructs the exact four unsigned big-endian integers, not just their boxes; MF_source adds beta once. Every raw/compressed hash, bit length, minimal encoding, marker monomial and physical support is checked. The complete native section12 constructs fresh canonical outer data matching section6 exactly. The free-chart endpoint in section5 constructs fresh original source witnesses, so the native sections5–8 apply. This establishes both interfaces rather than substituting a new tuple into an old calibration.

The historical radix67 specialization is separately supplied with the full parent finite recipe and little-endian payloads,295388 raw bytes. The new tuple is a different specialized polynomial and is not claimed identical to either radix67 or any older4177/4027 specialization.

4.9 Exact algebra, degree, and complete highest forms

The checker independently expands the center over Z[A,Ybar,k,U,b,c], using weights(4,2,1,5,4,2). It compares the division-free actual graph against(10) and the t-quotient identity. Put Def=Omegaˆ2-rˆ2GVˆ2. The weighted degrees of Omega and Def are

```
quotient_t:59,106; product2:59,102; product4:59,94.
```

For quotient_t the exact root defect is

```
C2^2-4G=U^2[4E(Rt-Qt)+Qt^2]-4E.
```

No positivity of Rt-Qt is required. The full three defect leaders are

```
16A^10 U^6 Ybar^10 b^2 k^8-16A^11 U^6 Ybar^8 b^2 k^8,
4A^10 U^6 Ybar^8 b^2 k^8,
8A^8 U^6 Ybar^8 b^2 k^8-48A^6 U^6 Ybar^12 b^2 k^8.
```

The complete output identity is

```
P=Q[Z0^2-2M Z0 Omega+M^2Def]+r^2M^2G-(Z0-M Omega)^2.
```

The independent degrees are $\deg(Q, M, G, V) = (72, 12, 46, 36)$. For transport bV we have $\deg Z_0 = 54$; for b^2, it is 22. The corresponding summand bounds are

$$(180, 197, 202, 70, 142), \quad (180, 197, 198, 70, 142), \quad (116, 165, 190, 70, 142).$$

In each case the defect summand uniquely dominates. Let $q_0 = \beta J$ and $k_0 = \eta + \zeta$. The complete collected top forms are

```
[P_quotient_t]_202 = 36864 q0^80 w^22 s^55 i^10 k0^32
                           (q0-F)^2(s-w),
[P_product2]_198 = 9216 q0^78 w^22 s^54 i^10 k0^32(q0-F)^2,
[P_product4]_190 = 18432 q0^74 w^18 s^52 i^10 k0^32
                           (q0-F)^2(w^2-6s^2).
```

They have198,99,198 terms respectively and are nonzero for each actual positive beta. All terms, not a selected numerical ray, are independently collected. Every gate is covered by an exact free-ring cut, all registers, inputs and ports are live, and both alternative operators at every gate are rejected:194,222,238 corruptions per tuple. Normal and optimized Python run the same checks, propagate their mode to child replays, and produce equal receipts. Relocated replay uses only this package.

5 A 201-variable natural quartic

All coordinates in this section range independently over $\mathbb{N} = \{0, 1, 2, \ldots\}$. The principal polynomial is the sum of the squares of 153 literal quadratic residuals. The defining residual list and its complete collected polynomial are supplied respectively in

`natural201/equation.json`
`natural201/expanded_polynomial.json`

It has 201 variables, exact degree 4, 6002 nonzero monomials and coefficient height 1860103. The inherited fixed-calculus calibration is unchanged: it has a natural zero if and only if the specified effectively presented ZFC calculus has a finite syntactic refutation. Its no-natural-zero sentence thus retains the inherited Con(ZFC) interpretation. This is not a consistency proof, an optimality claim, or a new end-to-end kernel formalization.

Profile	Variables	Rows	Degree	Terms	Height
Principal	201	153	4	6002	1860103
Affine parent	202	154	4	6000	1854571
Sparse sextic	165	117	6	6558	1860101
Smaller sextic	164	116	6	15774	1860101
Sparse octic	139	91	8	11873	1117699986
Smaller octic	138	90	8	16515	2933403776

Natural affine charts certified modulo source equations. Let $R = (R_0, \ldots, R_{m-1})$ be integral polynomials in n coordinates. Suppose a set S of coordinates has retained source equations

$$R_{j(v)} = X_v - f_v(X), \qquad f_v \in \mathbb{N}[X], \quad v \in S,$$

and an ordering in which every coordinate of S appearing in f_v is earlier than v. Let F be an integral affine automorphism, with integral affine inverse G, fixing the coordinates outside S. Suppose that for each $v \in S$ a certificate supplies

$$F_v = P_v + \sum_j a_{vj} R_j, \qquad P_v \in \mathbb{N}[X], \quad a_{vj} \in \mathbb{Z}. \tag{N1}$$

Then F and G give inverse bijections between the natural zero sets of R and $R \circ G$.

Indeed, at a natural zero x of R, equation (N1) proves $F_v(x) \geq 0$; the other coordinates are unchanged. Conversely, if y is a natural zero of $R \circ G$, put $z = G(y)$ in $\mathbb{Z}^n$. Coordinates outside S are already natural. The retained equations give $z_v = f_v(z)$, so induction in the stated source order proves every remaining coordinate nonnegative. The two affine inverse identities prove the inverse laws. Crucially, no claim is made that G preserves the entire natural orthant. An elementary integral row operation $R_i \leftarrow R_i + kR_j$, $i \neq j$, is reversed by subtraction and does not change the common zero set. Affine substitutions do not increase degree; applying the affine inverse proves preservation of maximum residual degree.

The literal chart. The source is the preceding pooled 202-coordinate natural system, with its full source closure and proof retained in `inherited/`. A positive syntax-index-budget accumulator, the twelve earlier syntax shears, and forty further literal affine shears yield the selected chart; twenty-four recorded elementary row shears are then applied. Complete replay inputs and both affine maps occur under `chart202/`. There are 33 changed coordinates. Twenty-nine final forward maps are coefficientwise nonnegative. The other four require only source rows $13, 17, 34, 75$, for coordinates $27, 31, 49, 93$ respectively. For example,

$$F_{31} = X_{27} + X_{31} - 16X_{25}, \qquad F_{93} = X_{93} - X_{92}.$$

Subtracting source row 17 from F_{31} gives the recorded polynomial with nonnegative coefficients. Source row 75 has the form $X_{93} - HX_{92}$, where H is coefficientwise nonnegative with constant 236; thus $F_{93} = (H-1)X_{92} + R_{75}$ is nonnegative on source zeros. All 33 original positive defining equations and their dependency order are checked from literal polynomials. The inherited Pell and power interpretations are used only after the original natural zero has been restored; no lost orientation is assumed away.

The new quadratic contraction. In the final 202-coordinate chart, before renumbering, the only rows containing X_{57} are

$$R_{42} = X_{57} - 2 - 6X_{48} - 4X_{48}^2, \qquad R_{44} = X_{59} - 4X_{57} - X_{58} - X_{58}^2.$$

The coordinated shear of X_{59} has removed the other incidences. Substitute $X_{57} = 2 + 6X_{48} + 4X_{48}^2$, and delete coordinate 57 and row 42. The remaining occurrence becomes

$$X_{59} - 8 - 24X_{48} - 16X_{48}^2 - X_{58} - X_{58}^2.$$

Every residual remains quadratic. The restoring polynomial is nonnegative on every natural retained assignment, and the deleted monic row forces precisely that value. Deletion and extension are therefore inverse natural zero-set maps. The resulting sum of 153 squares is the asserted 201-variable quartic. The literal collected coefficients establish exact degree, support and height. No lower-bound theorem is inferred from the finite search for further contractions in this particular row basis.

Signed monic contractions for the degree tradeoffs. There is a more general certificate which avoids requiring every restoring polynomial to have nonnegative coefficients. Start with a chart $C = U(R \circ G)$, where U is an invertible integral residual matrix. Repeatedly select a row $aY_v + h(Y_{\neq v})$, $a \in \{1, -1\}$, substitute $Y_v = -ah$ into the remaining rows and delete that row and coordinate. These operations give a polynomial extension E of the retained coordinates and a projection π, inverse to one another on the integer zero sets. This follows inductively because each deleted monic equation forces its restoring value, without division or an additional congruence.

Let D be the set of deleted source-coordinate indices. Assume the original system has positive acyclic defining equations for every coordinate in $S \cup D$. Given a natural retained zero y, extend to $E(y)$ over the integers and put $z = G(E(y))$. All original residuals vanish. Coordinates outside $S \cup D$ remain unchanged natural coordinates. Induction along the original positive definitions proves $z \in \mathbb{N}^n$. Equation (N1) then gives $F(z) = E(y) \in \mathbb{N}^n$. The inverse direction is projection of the natural chart image of an original natural zero. Hence the same maps are inverse bijections on natural zero sets, even when intermediate restoring formulas have negative coefficients.

The 165-, 164-, and 139-variable companions contract 37, 38, and 63 coordinates from the new chart; their certified restoration sets have 69, 70, and 83 coordinates. The 138-variable companion contracts 64 coordinates from the preceding-round chart and has a 74-coordinate restoration set. In the degree-eight profiles, the deleted original outer x is restored from its explicit positive equation $x = 1+\text{endpoint.ut}(1+\text{mul.8})$; no sign of a free integer is postulated. The equivalence is existential, not a new assertion with x held as an externally fixed parameter. Every contraction equation, substitution, restoration order and retained-index renumbering is supplied. The residual degrees are at most three or four, giving the displayed sextic and octic tradeoffs. They are not quartic-dimension improvements.

Current-source comparison and verification boundary. At the pinned current snapshot `ecb9a4b4dda617512952525d50f17343700aac52`, the authenticated source178 principal and sparse natural 203-variable quartics have 155 rows and respectively 8237 terms/height 799349232 and 8263 terms/height 207116893. The new 201-variable quartic improves all four measures against both. The source178 low-height alternative has 49682 terms and height 481920; it remains a height tradeoff. Literal current-source equations and a full independent recollection are included, so this is not a comparison with an obsolete summary.

The preceding principal had 202 variables, 154 rows and 9353 terms. The new counts are 201, 153 and 6002. Height rises from 1854167 to 1860103, an increase of 5936 (approximately 0.32%). Accordingly, we do not claim simultaneous improvement of every previous metric. The inherited low-height 202-variable form, with 16927 terms and height 416488, is preserved. The new 165-variable sextic strictly improves every displayed metric over the earlier 200-variable octic, which had 152 rows, 10392 terms and height 13654244.

The standard-library-only builder reconstructs all literal rows and coefficients. An independent exponent-count implementation verifies the affine inverse identities, positive representatives, original restoration DAGs, monic eliminations, every residual and every expanded coefficient. Negative controls reject corrupted representatives, gate orders, eliminated definitions and coefficients. Normal and optimized Python execution, byte-identical temporary reconstruction, LaTeX round-trips, current-source recollection and replay of the complete inherited finite construction closure are recorded. These are finite exact certificates accompanying the ordinary unbounded semantic and logical proofs, not a claim of a new kernel proof.

6 An even affine horizon: a quartic in 209 unrestricted integers

6.1 Result, comparison and exact source

The equation in `integer209/equation.json` is the sum of the squares of 149 literal quadratic residuals in 209 independently unrestricted integer variables. Its collected polynomial has degree 4, 5,486 nonzero terms and coefficient height 17592186044416=2^44. The maximum residual coefficient is 4194304=2^22. Every coefficient is supplied, with exact maps and an independent checker.

The immediate parent is the late Pell-growth210 equation at bounded snapshot `c8187221`. Its unchanged original archive is retained as `inherited/Source210.zip`, with SHA-256

```
33fecaaa728d5f38608b8b0930c09b08dbe9a01e03f6269f092ec6142d72a0ee
```

The exact parent stage, `equations/principal210/equation.json`, is copied byte-for-byte as `sources/parent210.json`. Its SHA-256 is

```
20a2108353a3756ed7d1ddfdc71695131dcf0b6d94039988d9565ac48e6e1f0a
```

The parent has 210 variables, 150 rows, 5,282 terms and height 2^40. This construction saves one variable and one row, at the cost of 204 additional terms and sixteen times the height. It is a dimension/coefficient tradeoff, not dominance. The separately verified 212-variable scaled-pair construction is unchanged and retained as a method control. No global minimum or historical-priority claim is made, and this bounded candidate is the final construction tested here.

The new ordinary proof, composed with the identified inherited calibration, establishes in ZFC that this polynomial has an integer zero exactly when there is a finite syntactic ZFC refutation. Negating gives nonsolvability iff Con(ZFC). It does not prove consistency, inconsistency or a zero. The argument is internal to ZFC, without a restriction to externally standard proof codes.

The credited incoming source supplies standalone Pell growth, its sharper constructor majorant, the affine radix, dyadic residue permutation, squarefree beta spacing, exact projections and all logical calibration. Those are not rediscoveries. The new contribution is the even-affine horizon together with the signed bootstrap, smaller-threshold estimates and changed cofinal parity construction needed to remove its private root. Every unchanged row is literally retained after one affine substitution and renumbering. No fresh Lean/Rocq build is claimed; the full original ordinary-proof package is retained, not represented as a complete compilable Lean dependency closure.

6.2 The literal change and its semantic scope

Let delta=2^19=524288. The incoming construction has the actual horizon

```
t=16+4h_old^2.
```

Replace it by

```
t=4h+2,            h independently unrestricted.          (1)
```

The old root is coordinate 178 of the frozen 210-coordinate source and occurs only in residual 125. Substitute 4h+2 for coordinate 0 in every other residual, delete row 125 and coordinate 178, and renumber the retained coordinates. Every residual stays quadratic. Exactly five retained rows change: the relevant front endpoint row, shared frame row, orientation row, standalone Pell product row and growth-divisibility row. The certificate gives their full old/new coefficient arrays, not just these descriptions.

The inherited reduction also commutes with this relaxation: throughout all six projection stages the old root stays private; the five radical directions and restoration polynomials avoid both t and the root; the final affine consequence uses only the local front rows 5 and 18. All 79 final invertible operations avoid the horizon row/root and never change t. Thus applying the semantic change to this authenticated optimized parent does not conceal a previous use of the old horizon in another retained condition.

This is a semantic zero-existence change, not a bijection between arbitrary old witness tuples: the old horizons are 0 modulo 4, the new horizons 2 modulo 4. The retained physical-coordinate map sends new h to actual t=4h+2 and leaves all other retained physical coordinates unchanged. On physical tuples with t=2 modulo 4, its inverse is h=(t-2)/4. The omitted old root is not reconstructed. The two directions of the consistency equivalence are proved below by arithmetic extraction and a fresh cofinal witness construction.

Keep the original standalone center 4t+1, with existing coordinates F and three private growth coordinates X,Z,u:

```
Z=tF,
X^2-16Z^2-8ZF=1,
F=delta*t*(2u+1).                                    (2)
```

No enlargement to center 8t+1 is required. Keep the even representative C, selector, affine radix and exact-power request

```
B*eta=C+F+1,   D_radix=2(C+F+1),   q=D_radix^x.          (3)
```

The last equality is proof notation for its retained quadratic certificate. All fifteen remaining requests, forty-five base/index/output ports, thirteen principal pairs, anchor comparison rows and normalized digit rows survive. The old square frame is retained exactly, including a harmless duplicate t atom left over from the obsolete growth request. Its34 actual affine atoms are fully reconstructed in `frame_certificate.json` before and after substitution. Thus its equation remains W=2+sum(atom^2), and every actual requested base, index and output has square less than W.

6.3 Soundness: recover the horizon before using positive growth

From (1), t is never 0 or -1 and |t|>=2. The last row of (2) alone gives

```
F!=0, F even, |F|>=delta*|t|>=2^20.                  (4)
```

No Pell classification at a positive growth center is used here. In particular the actual frame contains F and has W>=2+F^2>258 even when t or F is negative. The entire inherited signed C-anchor and orientation proof therefore applies before any sign recovery. Briefly, its nonnegative cap forces a nonzero anchor ordinate with absolute index|C|; squared Pell ordinate divisibility and the comparison congruences identify that index without assuming C positive. The other classified absolute indices are strictly smaller than|C|. Reduction of the pooled orientation modulo A-1 then gives a nonnegative residue smaller than |A-1|, forcing every principal pair to be its canonical signed pair. This uses only the retained actual frame bounds and cap, which are authenticated coefficient by coefficient. Consequently the inherited signed-power lemmas are available: a nonunit base forces a nonnegative index and the genuine integer power.

Retain y=F(2v+1), so |y|>=|F|. Because C and F are even, C+F+1 is odd; D_radix is nonzero even and has absolute value at least 2. Its power request first gives x>=0. If x=0, the last endpoint

```
x=1+[1+(t+1)y]ut
```

would make 1+(t+1)y divide -1. But t+1!=0 and

```
|1+(t+1)y|>=|t+1||y|-1>=|y|-1>1.
```

Thus x>=1 and |q|>=2. The q-pool is now genuinely nonunit. Its retained t and y+1 requests imply t>=0 and y+1>=0. Since t=2 modulo 4, t>=2. A negative y would be at most -|F|<-1, so

```
t>=2,              y>=|F|>0.                              (5)
```

This order is essential: positivity of t was not assumed to exclude x=0. The rejected odd-affine horizon allowed t=-1,x=0,q=1 and failed exactly here; `review/REJECTED_ODD_HORIZON.md` preserves that diagnostic.

Only now classify the standalone Pell equation at its positive odd center 4t+1. The growth estimate below gives |F|>t. If x=1, the first endpoint x=F+1+(y+1)u0 would make the nonzero F divisible by y+1 while |F|<=y<y+1, impossible. Thus x>=2. The last endpoint now has a positive multiplier and ut>=1, giving x>=2+(t+1)y. Hence |q+1|>=3, so its pool is legitimate. Its F+t request gives F+t>=0; together with|F|>t this forces F>0. The C request gives C>=0 and the already established anchor nondegeneracy gives C>0. Thus D_radix>0,q>0, and the first endpoint gives u0>=0 and x>=F+1. No postponed arithmetic sign is used as its own premise.

6.4 The original center and constructor bound work at t=2

Write X_n(a),Y_n(a) for the standard Pell sequences, with initial pair(1,0) and recurrence X_next=aX+(a^2-1)Y, Y_next=X+aY. For a>=2, every nonzero integer Pell ordinate has absolute value Y_n(a) for a positive integer n. The identities

```
Y_n(a)=n mod(a-1),
Y_(2n)(a)=2X_n(a)Y_n(a)
```

give v2(Y_n(a))=v2(n) when a is odd: X_n is always odd and Y_n is odd exactly when n is odd. Also Y_n(a)>=(2a-1)^(n-1).

For the actual square-shell constructors, the largest value on arguments at most M satisfies

```
E(M,M)+1=8M^2+14M+8<=8(M+1)^2.
```

At t>=2, every local parameter and every constant leaf 0 or 1 has value-plus-one at most t. The unchanged authenticated maximum constructor depth is 19. Induction therefore bounds every formula label strictly below

```
L=8^(delta-1)*t^delta.                                    (6)
```

This sharper bound needs only t>=2, not t>=16. It covers the same exact fourteen outputs and nine schema trees as the preserved source ledger.

Equation(2) is X^2-((4t+1)^2-1)F^2=1. Let |F|=Y_n(4t+1). Because 4t divides F, the first Pell congruence gives 4t|n. The odd multiplier in(2) and the exact valuation give v2(n)=19+v2(t). Hence n/(4t) is a positive integer of exact valuation 17, and n>=delta*t. Consequently

```
|F|>=(8t+1)^(delta*t-1)>=(8t)^(delta*t-1)>=L^t.          (7)
```

The exact ratio of the last lower bound to L^t is 8^(t-1)/t>=1. In particular |F|>t, as used only after (5) in the bootstrap. After F positivity has been recovered, (7) is the weighted-label separation budget F>=L^t.

Two other small-threshold arguments must be updated explicitly. Since t>=2, 8t+1>=17 and delta*t-1>=t, equation (7)'s first bound gives

```
F>=17^t>16^(t-1)>=lcm(1,...,t)^2.                    (8)
```

Thus the all-primes binomial selector remains valid at t=2. We do not reuse the false small-threshold assertion t^t>=16^t. The same strong Pell bound also gives F>2^32: delta*t-1>=8 and 17^8>2^32. From y>=F>t, t>=2 and x>=2+(t+1)y, R0=2+t+x+y+F<2x. If 2^s<=x<2^(s+1), then s>=32 and

```
2^23 log2(2x)<2^23(s+2)<2^s<=x.
```

The last inequality starts at s=32 and persists by induction. The inherited literal local-degree bound therefore gives R0^(2^23)<2^x<=p^(ex), so the same selected local congruences recover exact local rows.

6.5 Complete arithmetic extraction and logical calibration

The unchanged C-cap makes every other nonnegative nominal index strictly smaller than C. In particular F+t<C, all ordinary upper arguments are at most C, and every needed lower digit index is at most x. Equation (3) gives

```
D_radix=2(C+F+1)>2C,     q=D_radix^x>(2C)^x.
```

Every binomial coefficient in a retained digit is therefore at most C^x, which is less than q/2^x. The geometric-series tail is less than its digit denominator divided by 32 because x>=F+1>2^32. The common normalized remainder norm bounds the competing remainder in the centered half interval. Their congruence thus forces equality. For the first digit, the nonzero selector quotient gives |B|<=C+F+1=D_radix/2<q/2. Its true positive binomial coefficient is less than q/32. The digit congruence yields

```
B=binom(F+t,t)>0.
```

The unchanged B-base power gives N=B^x, while B|D_radix from (3) gives N|q. Thus all ten digit/range predicates, every local interval, and all local precision assumptions are authenticated exactly as in the incoming source. The all-primes selector (8) covers each row and every prime of B in both soundness and completeness.

For labels bounded by L, equation (7) supplies the source's sharp product separation theorem, including equality F=L^t. Unequal factor count s are impossible because F+1>L^t. At equal count m<=t, each positive elementary symmetric coefficient lies in[1,L^m] subset[1,F], so differences have absolute value less than F. The least nonzero coefficient cannot disappear on evaluation at F. Thus label multisets coincide. Zero labels contribute units and are omitted. The unchanged boundary and positive-accumulator arguments recover a finite single-conclusion Horn flow; the inherited closure proof extracts a refutation even if the flow has cycles. In particular any exclusion of y=F is made after local recovery, not before.

The literal local arithmetic uses the square-shell presentation. First apply the retained injective square-shell/original-C222 recoding, which carries finite refutations in both directions. This

unchanged recoding precedes the logical calibration; the shell labels are not silently identified with the original numerical codes. The retained named endpoints are `KEndpoint.c222_iff_zf_refutation` and `ConstructibleChoice.zf_refutation_iff_zfc_refutation`. They compose the literal arithmetic extraction with a finite ZFC refutation. No machine execution or Jones-polynomial specialization is substituted for these calibrated endpoints. This proves soundness.

6.6 Cofinal completeness with the changed dyadic bit

Start with a finite C222 refutation and all its finite local parameters, proof length and padding. Choose a>=11 sufficiently large and set

```
t=2^(2a)+2,        h=2^(2a-2).                    (9)
```

These horizons are cofinal and satisfy (1). They have v2(t)=1 and binary bit 20 equal to zero, since their only nonzero bits are 1 and 2a>=22. Put n0=delta*t, y0=Y_n0(4t+1), a0=X_n0(4t+1). The congruence gives t|y0, and the valuation gives v2(y0)=20. Hence

```
b0=y0/(delta*t)
```

is an odd integer. Also a0=1 modulo 4 because n0 is even and the abscissas at odd center are odd.

Use the inherited normalized Pell permutation lemma: if a0=1 modulo 4, s->Y_s(a0) induces a permutation modulo every 2^r. Here is its unbounded proof. For H=2^r, the valuation gives v2(Y_H)=r. The norm and X_H=1 modulo 4 give X_H=1 modulo 2^(2r+2). Addition then yields

```
Y_(s+2^r)(a0)-Y_s(a0)=2^r mod2^(r+1).
```

The two lifts of each index class have opposite next output bit. Induction from parity proves bijectivity and periodicity; adding positive multiples of 2^r gives cofinal indices in a selected class.

Choose b>22 with 2^b>t. Since (t/2)b0 is odd, the permutation supplies a positive odd indexs, as large as desired, satisfying

```
(t/2)*b0*Y_s(a0)=1 mod2^(b-20).
```

Set n=n0*s and

```
F=y0Y_s(a0)=Y_n(4t+1),
X=X_n(4t+1), Z=tF, u=(F/(delta*t)-1)/2.
```

All values are integral, F is positive and arbitrarily large, and every row of (2) holds. These choices give

```
F=2^20 mod2^b.                                  (10)
```

Below bit b, F has only bit 20, which is absent from t; at or above bit b, t has no nonzero bits. Therefore the binary supports of F and t are disjoint. The identity v2(m!)=m-s2(m) gives

```
v2 binom(F+t,t)=s2(F)+s2(t)-s2(F+t)=0.
```

Thus B is odd. This changes the incoming bit 23 construction to bit 20, rather than silently reusing its old t=16 modulo 2^26 horizon.

Choose an arbitrarily large positive odd w divisible by every odd prime <= t, and put y=Fw. Every prime<=t divides y. If 1+iy and 1+jy had a common prime p for 1<=i<j<=t+1, then p is coprime to y and divides j-i, so p<=t; but then p|y, a contradiction. Hence all beta moduli are pairwise coprime. This uses prime support, not the false assertion t!|y with fixed v2(F)=20. Choose w large enough to dominate all pending-state values and then choose a positive beta code x arbitrarily large in its CRT class. The odd-linear beta parameter is(w-1)/2. Define N=B^x only after x is fixed; N is odd.

The generalized factorial-free selector (8) assigns the independent CRT representatives modulo N^2. Choose the eight ordinary representatives positive first. BecauseN^2 is odd, chooseC even, positive and cofinally large in its prescribed class, exceeding all other requested indices and representatives. The selector congruenceB|C+F+1 remains true. Only then setD_radix, q and their exact powers. We haveD_radix>2C and N|q, so the same digit bounds supply all normalized quotients and remainders.

Choose the shared square frame after all these arithmetic outputs are fixed. Its canonical positive principal pairs, private even-C anchor comparisons, cap slack and orientation quotient are completed exactly by the retained source construction. All their hypotheses have just been checked, without a cofinal-square horizon. Evaluate every ordinary circuit wire, substitute the new h in(1), omit the old root slot and renumber. This constructs an integer zero from every finite refutation and completes the ordinary equivalence.

6.7 Exact verification and portability

`semantic_certificate.json` pins the whole original210 equation, every changed row, the exact substitution and the private root/row deletion. `frame_certificate.json` independently reconstructs all 34 actual square atoms and covers all 45 actual power ports. `coordinate_map.json` gives all retained coordinates, rows and exact renumbering. The independent checker imports neither producer nor producer algebra library. It compares every row, verifies the inverse affine polynomial identities on retained physical tuples, recollects every quartic coefficient and parses all 298 plain/TeX residual displays back into polynomials. Actual-data corruption controls must fail with and without Python -O.

Verification is read-only unless an explicit receipt path is supplied. Every verification rebuilds from a fresh relocated copy and byte-compares all generated JSON. The optional parent replay authenticates the original archive, the immediate principal210 stage, its inherited source 173 closure, all source 210 stages and the full original package verifier. New certificate paths are relative; historical strings inside the unchanged original archive are provenance, never paths used to find new inputs. No root search, remote write or proof-assistant rebuild is involved. Finite checks certify literal algebra and source binding, not the unbounded consistency theorem by testing.

7 Reproduction and the logical boundary

The direct constructions supply every residual and every collected coefficient of the final sum of squares. The compact constructions supply every live circuit gate, all actual fixed integer payloads and their finite reconstruction recipes. Powers, fractions, factorials, recurrences and inequalities in semantic proofs are not hidden primitives or extra quantified variables inside a final polynomial.

Ordinary unbounded mathematics establishes sign recovery, nonzero resultant denominators, cofinal choices and constructive maps. Independent finite checkers bind these proofs to literal rows,

inverse maps, all leading terms, coefficient bytes and actual corruption controls. Finite sample identities supplement, not replace, their unbounded proofs. The portable release replay runs in fresh copies under normal and optimized Python, propagates optimized mode to child checkers, and preserves sealed bytes. The actual extracted ZIP is replayed separately before delivery.

The logical calibration remains the specified inherited numerical calculus or native, compiler and refutation theorem. No transitive formal development is claimed to have been rebuilt into a new proof-assistant theorem for every printed coefficient. The independent reviews are separate checks within an AI-assisted workflow, not outside human endorsements. No upstream repository is modified, and no literature-wide priority or global simplicity lower bound is asserted.

The low-padding2963 language and four-number tuple differ from the older indexed943/radix67 specialization. Physical positions with zero weight remain part of the native proof. Each complete tuple, selector map, native proof and finite controller is authenticated separately; neither scalar bounds nor weighted-array equality license exchanging ports or certificates. Storage byte order and compression metadata are distinguished from the mathematical integers. Any documented inherited gzip-header portability mismatch is not relabeled as a successful compressed-byte rebuild.

A The complete historical radix67 tuple

In this appendix K denotes the port K_0 in the new equations. Every symbol introduced here is a fixed constant, independent of the ten quantified variables. The formulas and 436-character seed are the retained indexed BFS943 radix67 specification; they reconstruct the exact supplied payloads, with $\beta = 2^{590069} - 1$.

The four fixed integers. Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\rho = 2^{67}, \quad B = \rho^{8807}, \quad \beta = B - 1, \quad G_m(x) = \sum_{j=0}^{m-1} x^j.$$

Define

$$\begin{aligned}\mathcal{E} = {} & \{0, \dots, 942\} \cup \{983, \dots, 992\} \cup \{1003, \dots, 1011\} \\ & \cup \{1013, \dots, 1031\} \cup \{1063, 1064, 1095, 2195, 3296\}.\end{aligned}$$

Then

$$\begin{aligned}M_C &= B - 1 - \sum_{e \in \mathcal{E} \setminus \{1\}} \rho^e - 2\rho^{1063}, \\ M_F &= B - 1 + 4 + (2^{51} - 2)(\rho^{4393} + \rho^{5495}) \\ &\quad + \rho^{1013} G_{10}(\rho) + \rho^{1065} G_9(\rho) + \rho^{2195} + \rho^{3296} \\ &\quad + (2^{65} - 1)\rho^{8792} G_{13}(\rho) + (2^{19} - 1)\rho^{8805}, \\ K &= \rho^{30} + \rho^{62} + \rho^{1100} + \rho^{2201} + \rho^{8849} \\ &\quad + (\rho^{4393} + \rho^{5495})\Phi(1/\rho),\end{aligned}$$

where

$$\Phi(x) = (1+2^{51})G_{943}(x) - 2^{10}\sum_{i=0}^{942} v(t_i)x^i + \sum_{(b,e,m)\in\mathcal{C}} 2^b x^e G_m(2x),$$

$$\begin{aligned}\mathcal{C} = \{&(10,1023,8),(18,1013,9),(27,1003,8),\\ &(35,983,9),(44,1031,1),(45,1022,1),(46,1011,1),\\ &(47,992,1),(48,1095,1),(49,2195,1),(50,3296,1)\}.\end{aligned}$$

The rational intermediate in K cancels exactly; all four results are positive integers.

The 943 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. Let

$$h(q,b) = 2(q+1)+b, \quad \mathcal{R} = \{u_j : d_j = 1,\ 0 \le j < 430\},$$
$$\mathcal{L} = \{u_j : d_j = 0,\ 0 \le j < 430\},$$
$$r_q = 6 + |\mathcal{R} \cap \{0,\dots,q-1\}|, \quad l_q = 205 + |\mathcal{L} \cap \{0,\dots,q-1\}|.$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 942$; all index ranges are fixed:

index i	tile $t_i = (w,n,e,s)$
$0, 1$	$(2,435,3,1),\ (5,0,0,0)$
$j+2\ (j = 2q+b < 430)$	$((1-d_j)l_{u_j}, h(q,b), d_j r_{u_j}, v_j)$
$432+b\ (b=0,1)$	$(0, h(215,b), 0, h(215,b))$
$434+b\ (b=0,1)$	$(0,b,0,b)$
$424+2r_q+b\ (q\in\mathcal{R}, b=0,1)$	$(r_q,b,0,h(q,b))$
$424+2l_q+b\ (q\in\mathcal{L}, b=0,1)$	$(0,b,l_q,h(q,b))$
$924+j\ (0\le j<4)$	$(0, j+430\lfloor j/2\rfloor, 0, 438)$
928	$(2,434,2,0)$
929	$(3,436,4,4)$
930	$(4,436,4,0)$
$497+t\ (t=434,435,436)$	$(1,438,1,t)$
$934+3a+b-6\lfloor a/4\rfloor$ $a\in\{0,1,4\},\ b=0,1,2$	$(a,437,b,437)$

For $t=(w,n,e,s)$ put

$$v(t) = w + 2^8 n + 2^{17} e + 2^{25} s + 127\,2^{34}.$$

A closed decoder for the 430 actions. Put $D_b(A,k) = \left\lfloor A/b^k \right\rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b\rfloor$. For $0 \le q < 215$, $b = 0,1$, $j = 2q+b$:

$$N = \mathcal{A} \bmod 2^{636}, \quad T_0 = \left\lfloor \mathcal{A}/2^{636} \right\rfloor \bmod 2^{1675},$$
$$E_0 = \left\lfloor \mathcal{A}/2^{2311} \right\rfloor,$$
$$\nu_j = D_2(N,j), \quad \sigma_j = \sum_{k=0}^{j-1} \nu_k,$$
$$m_q = D_2(N, 430+q), \quad \tau_q = \sum_{k=0}^{q-1} m_k,$$
$$F_q = 14(1-m_q) + m_q D_{16}(E_0, \tau_q),$$
$$u_j = \nu_j(1+\sigma_j) + (1-\nu_j)D_{216}(T_0, j-\sigma_j),$$
$$v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b+1).$$

Empty sums are zero. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$. The alphabet, in order of digit values $0, \dots, 61$, is

```
0123456789ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz.
```

Concatenate the following 436 characters as digits $e_0, \dots, e_{435}$; put $\mathcal{A} = \sum_{j=0}^{435} e_j 62^{435-j}$.

```
BxEnhvy6NipzkWmROo0GsA3uK3twWLxscLq20UTumy5Cdu461LwD6g7RDNzF8Y
iYDFTpB8ehAypd7d6OsOL4APamnJeNXQypUezCmW9n4oXP6XfgmiWuAWZ8ZJ7U
OP7gu4kVdlGZqZHULQMar1dXNeJW4i7oNVI8usp8BlovjjZbwPCovZI328fRUq
4q92R81s5GjWQaQfGhkq9UYMK3rRMaBmsOIBkJqc2cj2VRIZ6mjzkBIOkW9SjZ
FzeA6J5u4j3sBL3SdgwXzM9mHvQivu8EJ1TBTpRhHDqZIbmYgwQ8VqTAFotl6F
We1YC554GKrVHsA8oUGKSA6F6k5KREBSD0Tgd7bFALY84LLbpw574BQyvdoHrl
0Oe35uwDf15DpT6VJWZBjSbHPpPAINYTte0eUxnTFTKBlGj8Am6sb1gLldh8WZ
8P
```

B The complete pinned low-padding2963 tuple

Here K denotes the fixed port K_0. No symbol in this appendix is another unknown. **The four fixed integers.** Every index, set, sum, product and floor below defines a constant, not another unknown. Put

$$\rho = 2^{47}, \quad B = \rho^{2963}, \quad \beta = B - 1,$$
$$G_m(x) = \sum_{j=0}^{m-1} x^j, \quad \Lambda = \rho^{1956}.$$

$$\begin{aligned} M_C &= \beta - \rho^{977} \\ &\quad - G_{942}(\rho) - \Psi(\rho, 1), \\ M_F &= \beta + 4G_{917}(\rho) + (2^{44} - 2)\Lambda \\ &\quad + \rho^{960} G_9(\rho) + \rho^{987} G_8(\rho) + \rho^{1005}, \\ K &= \rho^{18} + \rho^{1005} + \rho^{2972} \\ &\quad + \Lambda\Phi(1/\rho), \end{aligned}$$

where

$$\begin{aligned} \Phi(x) &= \sum_{i=0}^{941} [1 + 2^{10}(1 + w_i + 2^8 n_i \\ &\quad + 2^{17} p_i + 2^{25} s_i)] x^i + 2^{10}\Psi(x, 2), \\ \Psi(x, t) &= x^{969}(x^9 + t^{17}) G_8(tx) \\ &\quad + t^8 x^{942}(x^{18} + t^{17}) G_9(tx) \\ &\quad + (2 - t)x^{1005}. \end{aligned}$$

The rational intermediate in K cancels exactly; all four results are positive integers.

The 942 indexed fixed tiles. Use the actions u_j, v_j, d_j decoded below. For $d = 0, 1$, let

$$h(q, b) = 2q + b + 2, \quad \mathcal{S}_d = \{u_j : d_j = d,\ 0 \le j < 430\},$$
$$r_{d,q} = 205 - 199d + |\mathcal{S}_d \cap \{0, \dots, q - 1\}|.$$

An interval ending at -1 is empty. The table defines t_i for $0 \le i \le 941$; all index ranges are fixed:

index i	tile $t_i = (w_i, n_i, p_i, s_i)$
0	$(2, 435, 3, 1)$
$j+1$ $(j = 2q+b < 430)$	$((1-d_j)r_{0,u_j}, h(q,b), d_j r_{1,u_j}, v_j)$
$433 - 2a + b$ $(a, b = 0, 1)$	$(0, 432a+b, 0, 432a+b)$
$\substack{423+2r_{d,q}+b \\ q\in\mathcal{S}_d,\ d,b=0,1}$	$(dr_{d,q}, b, (1-d)r_{d,q}, h(q,b))$
$923 + 2a + b$ $(a, b = 0, 1)$	$(0, 432a+b, 0, 438)$
927	$(2, 434, 2, 0)$
928	$(3, 436, 4, 4)$
929	$(4, 436, 4, 0)$
$496 + t$ $(t = 434, 435, 436)$	$(1, 438, 1, t)$
$933 + 3a + b$ $(a, b = 0, 1, 2)$	$(a^2, 437, b, 437)$

A closed decoder for the 430 actions. Put $D_b(A,k) = \left\lfloor A/b^k \right\rfloor \bmod b$, where $a \bmod b = a - b\lfloor a/b \rfloor$. For $0 \le q < 215$, $b = 0, 1$, $j = 2q + b$:

$$N = \mathcal{A} \bmod 2^{636}, \quad T_0 = \left\lfloor \mathcal{A}/2^{636} \right\rfloor \bmod 2^{1675},$$
$$E_0 = \left\lfloor \mathcal{A}/2^{2311} \right\rfloor,$$
$$\nu_j = D_2(N, j), \quad \sigma_j = \sum_{k=0}^{j-1} \nu_k,$$
$$m_q = D_2(N, 430+q), \quad \tau_q = \sum_{k=0}^{q-1} m_k,$$
$$F_q = 14 + m_q(D_{16}(E_0, \tau_q) - 14),$$
$$u_j = \nu_j(1+\sigma_j) + (1-\nu_j)D_{216}(T_0, j-\sigma_j),$$
$$v_j = D_2(F_q, 2b), \quad d_j = D_2(F_q, 2b+1).$$

Empty sums are zero. Here $(v_0, d_0, u_0) = (1, 1, 1)$; the constructor tile uses this actual first target, including its state renaming.

The integer $\mathcal{A}$. The alphabet, in order of digit values $0, \ldots, 61$, is

```
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```

Concatenate the following 436 characters as digits $e_0, \ldots, e_{435}$; put $\mathcal{A} = \sum_{j=0}^{435} e_j 62^{435-j}$.

```
BxEnhvy6NipzkWmROo0GsA3uK3twWLxscLq20UTumy5Cdu461LwD6g7RDNzF8Y
iYDFTpB8ehAypd7d6OsOL4APamnJeNXQypUezCmW9n4oXP6XfgmiWuAWZ8ZJ7U
OP7gu4kVdlGZqZHULQMar1dXNeJW4i7oNVI8usp8BlovjjZbwPCovZI328fRUq
4q92R81s5GjWQaQfGhkq9UYMK3rRMaBmsOIBkJqc2cj2VRIZ6mjzkBIOkW9SjZ
FzeA6J5u4j3sBL3SdgwXzM9mHvQivu8EJ1TBTpRhHDqZIbmYgwQ8VqTAFot16F
We1YC554GKrVHsA8oUGKSA6F6k5KREBSDOTgd7bFALY84LLbpw574BQyvdoHrl
0Oe35uwDf15DpT6VJWZBjSbHPpPAINYTteOeUxnTFTKBlGj8Am6sb1gLldh8WZ
8P
```